

Recent Advancements in Biotechnology: Genetic Engineering (CRISPR and Gene Therapy)

Abstract

Genetic engineering has transformed biotechnology through precise modification of DNA. CRISPR-Cas9 and gene therapy have enabled treatment of inherited diseases, improved crop development, and accelerated biomedical research. This report reviews the principles, recent advancements, applications, benefits, challenges, ethical concerns, future scope, and concludes with APA references.

Introduction

Genetic engineering involves altering an organism's genetic material to achieve desirable traits. CRISPR-Cas9 is a powerful genome-editing tool that allows accurate DNA modification. Gene therapy introduces, removes, or repairs genes to treat disease. Recent advances include base editing, prime editing, improved delivery systems, AI-assisted guide RNA design, and successful clinical trials for inherited blood disorders, eye diseases, and cancers. Applications extend to medicine, agriculture, industry, and environmental biotechnology. Benefits include precision, faster development, improved health outcomes, and sustainable production. Challenges include off-target effects, ethical issues, regulatory oversight, cost, and equitable access. The future of genetic engineering is expected to include personalized medicine, safer editing technologies, and wider therapeutic use. Genetic engineering involves altering an organism's genetic material to achieve desirable traits. CRISPR-Cas9 is a powerful genome-editing tool that allows accurate DNA modification. Gene therapy introduces, removes, or repairs genes to treat disease. Recent advances include base editing, prime editing, improved delivery systems, AI-assisted guide RNA design, and successful clinical trials for inherited blood disorders, eye diseases, and cancers. Applications extend to medicine, agriculture, industry, and environmental biotechnology. Benefits include precision, faster development, improved health outcomes, and sustainable production. Challenges include off-target effects, ethical issues, regulatory oversight, cost, and equitable access. The future of genetic engineering is expected to include personalized medicine, safer editing technologies, and wider therapeutic use. Genetic engineering involves altering an organism's genetic material to achieve desirable traits. CRISPR-Cas9 is a powerful genome-editing tool that allows accurate DNA modification. Gene therapy introduces, removes, or repairs genes to treat disease. Recent advances include base editing, prime editing, improved delivery systems, AI-assisted guide RNA design, and successful clinical trials for inherited blood disorders, eye diseases, and cancers. Applications extend to medicine, agriculture, industry, and environmental biotechnology. Benefits include precision, faster development, improved health outcomes, and sustainable production. Challenges include off-target effects, ethical issues, regulatory oversight, cost, and equitable access. The future of genetic engineering is expected to include personalized medicine, safer editing technologies, and wider therapeutic use.

CRISPR Technology

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Recent Advancements

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Applications

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Advantages

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Challenges and Ethical Issues

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Future Scope

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Conclusion

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APA References

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